Correction of Lexical and Grammatical Errors in MT Results

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1. Introduction

Recently, Machine Translation (MT) has made significant achievements. However its quality has been known to be very poor, and has not reached users' satisfactory level. With the rapid progress of Internet technology, international conversation becomes more frequent. The necessary of improvement of MT performance proves to be even more serious. In [1], Schafer established a special error typology after gathering examples of errors typically corrected by post-editors in their various MT projects. In their classification, the translation error classification includes lexical errors (e.g., terminology, verb, noun), syntactic errors (e.g., word order and clause analysis) and grammatical errors (e.g., article and tense).

Most of the traditional researches on error correction have been using manually written rules. However writing a complete correction rule set is a time-consuming work that also needs the help of linguists. Methods for correcting errors using statistical approach have also been presented in [2, 3]. Atwell [2] detected errors as sequences with low probability of part of speech. Bigert and Knutsson [3] used a similar method, in which they compared new text to known correct text. Training a machine learning algorithm to detect errors has been tried in some researches. Mangu and Brill [4] used machine learning to detect when one word has been confused with another. Izumi et al. [5] attempted to detect errors by training a Maximum Entropy (ME) model. These researches discussed invariably required not only large amounts of training corpus but also manual annotation of errors, which is time consuming work with high cost.

Web-based models whose parameters of the model (web counts of queries) can be obtained with the help of WWW search engine Google, have been applied in a wide range of natural language processing. We have solved the countability of English compound noun using such models [6]. In this paper, we introduce an approach to correct lexical and grammatical errors in translation results to improve the quality of MT. In our correction algorithms, simple and viable web-based models are proposed to detect expression errors and select correct target expression from generated candidates. The approach is evaluated on erroneously selected articles and preposition errors. We experimentally showed that our approach that could save a lot of work of collecting large amounts of training corpus and manually annotating for training a machine learning model, could perform the promising results.

2. Our Approach for correcting errors

2.1. Web-based Models

In a translation result, we define C_1 as an expression (a phrase or word) with a suspected error. Our research goal is to detect whether there is any error in C_1 and correct the error in the further. $P=\{C_i| i >=1\}$ is a set of expression candidates, in which C_i stands a possible correct expression.

The Internet is a rich source of data for natural language processing. In our research work, we view WWW as a large expression dictionary and assume that the expression C with the largest occurrence probability is most likely correct given a certain context. We use Formula (1) as follow to describe C.

$$C = \arg\max \Pr(Ci \mid context) \quad (i \geq 1) \quad (1)$$

$\Pr(Ci \mid context)$ is the occurrence probability of C_i when given the *context*. We then use Maximum Likelihood Estimation (MLE) for a conversion from probability to frequency calculating as Formula (2).

$$\Pr(Ci \mid context) \approx \frac{\Pr(Ci, context)}{\Pr(context)} = \frac{frequency(Ci, context)}{frequency(context)} \quad (2)$$

With the help of a WWW search engine Google we can obtain web frequency values (web counts of queries).

Querying in WWW adds noise to the data, we certainly lose some precision compared to supervised statistical models, but we assume that the size of the WWW will compensate the rough queries. Keller and Lapata [7] showed the evidence of the reliability of the web counts for natural language processing. Although they also experimentally showed that web-based approach could overcome data sparseness for bi-grams, the problem still exists in our experiments. Therefore, when the web count returned is zero, we smooth zero by adding it to 0.01.

2.2. Processing Flow

Figure 1 gives an overview of our processing flow for correcting errors in translation results.

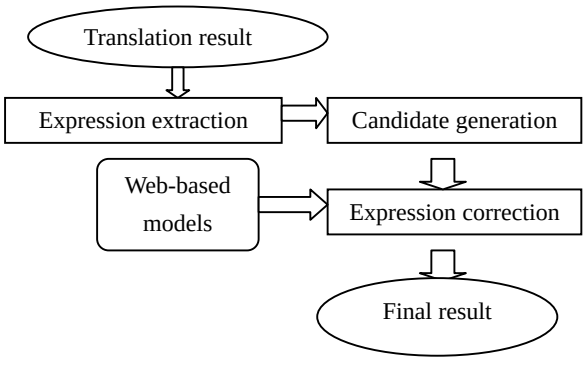

Figure 1. Processing Flow for Error Correction

For a translation result, we firstly need to extract the expression to process in the further. In the next step, we generate the other candidates based on the extracted expression. Then using the web-based model explained in section 2.1, we can get the final corrected result that has the largest probability estimated by the models.

3. Experiments for Performance Evaluation

We evaluate our ecorrection approach on erroneously selected preposition errors (lexical errors) and articles (grammatical errors) respectively.

3.1. Preposition Errors

A preposition describes a relationship between other words in a sentence. Preposition errors are commonly seen in translation results. For example, the translation result "#[1]Since he met his new girlfriend, Juan never seems to be in home" uses a preposition "in" when there should be "at". It is difficult to devise a complete set of rules for determining all the prepositions given certain contexts.

We evaluate our approach explained in section 2 on preposition errors using the following correction algorithm presented in Figure 2.

1.Extracting the preposition in a sentence and its context(l, r) using "Apple Pie Parser".
2.Generating the other preposition candidates $\{P_1, P_{2...}P_n\}$ using web- based knowledge acquisition.
3.Output argmax($Pr(P_1|Context), Pr(P_2|Context) \ldots Pr(P_n|Context)$)
4.Replace the preposition with argmax( ).

Figure 2. Correction Algorithm For Article Errors

In the algorithm, we use "Apple Pie Parser", which is a bottom-up probabilistic chart parser to extract preposition and its context. Context(l, r) means the context of the extracted preposition includes the l words left to the preposition and r words right to the preposition. We generate preposition candidates using web-based knowledge acquisition. Using Google operator "*" (In Google "* query", "*" can stand for any word in a search term.), we can acquire several possible prepositions with a certain context.

Table 1. Correction Accuracy

Context	Recall (%)	Precision (%)	F-score (%)
(1, 1)	81.2	83.7	82.4
(0, 1)	76.1	78.6	77.3
(2, 1)	78.5	82.2	80.3

We selected 200 sentences just only with preposition errors and manually generated their candidates analyzing the first 100 pages returned by Google for a "* query". We use Recall and Precision to evaluate the performance of our correction approach. Recall is the ratio of the number of preposition errors corrected accurately (A) against the total number of preposition errors (AB) while Precision is the ratio of A against the total number of preposition errors corrected (AC). We also introduced F-score when we need to consider the Recall and Precision at the same time. Table 1 shows the performances of our approach given 3 different contexts: (1, 1), (0, 1), (2, 1).

3.2. Article Errors

In article error correction, we consider article selection errors and singular/plural form errors together. For example, for the translation result "#It could be a growing pain", our research goal is to automatically correct the phrase "#a growing pain" to "growing pains", for *growing pains* is a plural only compound noun which always occurs in plural form and there should be no article in this sentence either.

We use a similar algorithm to that described in Figure 2, in which we generate the article form candidates using an article form table defined beforehand. The table consists 5 article forms (e.g., "a/an" and singular noun, "the" and singular noun).

For 1,000 sentences just only with article errors, when given four different contexts: (1, 1), (2, 0), (2,1) and (2, 2), the correcting precisions are 85.7%, 83.6.8%, 86.2% and 79.1% respectively.

4. Conclusion and Future Work

In this paper, we explained how to correct errors in translation results in order to improve the performance of MT. We experimentally showed that our approach based on the simple models could perform the promising results evaluating on two different problems of grammatical and lexical errors. Though we have not yet done any comparison experiments on the same test data with the early researches, we can achieve such promising accuracy based on the simple and unsupervised model. It used just only context frequency and co-occurrence frequency as features much less than those used in the models proposed in the literatures [2, 3, 4, 5].

As a part of our future work, we would also like to combine some general rules with our approach to improve the correct rate. Using WWW is an exciting direction for NLP, but the web-based methods invariably introduce noise in the resulting frequency data. Our next step is aiming at evaluating the Internet resource by distinguishing the useful and noise data.

References

[1] F. SCHÄFER, "MT-post-editing: How to shed light on the "unknown task" - experiments made at SAP", *Controlled Language Application Workshop*, page: 133-140. 2003.

[2] E. S. Atwell, "How to detect grammatical errors in a text without parsing it", *In Proc. of the 3rd EACL*, page: 38-45, Copenhagen, Denmark. 1987.

[3] J. Bigert and O. Knutsson, "Robust error detection: A hybrid approach combining unsupervised error detection and linguistic knowledge", *In Proceedings of Romand Robust Methods in Analysis of Natural language Data*, page: 10-19, 2002.

[4] L. Mangu and E. Brill. "Automatic rule acquisition for spelling correction". *In Proceedings of the 14th International Conference on Machine Learning*, page:187–194, 1997.

[5] E. Izumi, et al, "Automatic error detection in the Japanese learners' English spoken data". *In Companion Volume to the Proceedings of ACL '03*, page: 145–148, 2003.

[6] J. Peng and K.Araki, ""Detecting the Countability of English Compound Nouns Using Web-based Models", IJCNLP-05, 2005 (to be appeared)

[7] F. Keller, and M. Lapata, "Using the web to obtain frequencies for unseen bigrams", *Computational Linguistics* 29, 3, 459-484. 2003.

[1] "#" means that the text is grammatically wrong.